Extended sign-kill for $\widehat{\lambda} = (3, 2)$: the E_j^- chain at $j \leq q - 2$ via three-branch support

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Abstract

Let $\lambda = (3, 2, 1^q) \vdash n = q + 5$ with $q \geq 3$, $\ell = q + 2$, and $B = B^{(\lambda)}_{\ell+1} V_\lambda = R'_{\ell+1} \cdots R'_n V_\lambda$. We prove the "$\subseteq$" direction of the extended sign-kill conjecture for this family:

$$B \;\subseteq\; \bigcap_{j=1}^{q-2} E_j^- \,|_{V_\lambda},$$

matching the predicted threshold $\tau(\lambda) = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 = q - 2$. We further prove sharpness at the boundary: $B \not\subseteq E^-_{q-1}\,|_{V_\lambda}$.

The proof rests on the support-rule (May-14 fourth pass): if every SYT in $\mathrm{supp}(B)$ has j and $j{+}1$ in the same column, then $B \subseteq E_j^-$. We extract the support description from the May-20 three-branch reduction $B = (T_{n-3}{+}1)(T_{n-2}{+}1)\left[\Phi_A(B^{(\mu_A)} V_{\mu_A}) + \Phi_B(B^{(\mu_B)} V_{\mu_B})\right]$ together with column-1 identity-prefix bounds on the supports of the two inner pieces: the $(2, 2, 1^{n-6})$ piece is bounded by the May-14 second-pass paper, and the hook $(3, 1^{n-5})$ piece is bounded by a new general *hook support lemma* proved here by induction on the arm length k via the May-13 shift-lemma reduction.

Combined with the universal $B \subseteq E_\ell^+$ (May-13 evening Theorem A), this gives a clean four-piece eigenspace picture for the $\widehat{\lambda} = (3, 2)$ family: B sits in $E_\ell^+ \cap \bigcap_{j \leq q-2} E_j^-$, with both bounds sharp. This closes the $\widehat{\lambda} = (3, 2)$ row of the May-14 evening conjecture table in the containment direction.

1 Setup

We use the conventions of the May-13–May-14 series. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \ldots, T_{n-1}$ and quadratic relation $T_j^2 = (q-1)T_j + q$. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$.

For $1 \leq j \leq n - 1$, set $S_j := T_j + 1$. Write

$$E_j^+ \;:=\; \ker(T_j - q)\,|_{V_\lambda} = \; \mathrm{Im}(S_j\,|_{V_\lambda}), \qquad E_j^- \;:=\; \ker(S_j)\,|_{V_\lambda}\,.$$

For $k \geq 1$, let $R'_k := S_{k-1} S_{k-2} \cdots S_1$ with the convention $R'_1 = 1$. For $\lambda \vdash n$ with $\ell = \ell(\lambda)$, set

$$\Omega \;=\; \Omega^{(\lambda)} \;:=\; R'_{\ell+1} R'_{\ell+2} \cdots R'_n, \qquad B \;:=\; \Omega \cdot V_\lambda \;=\; B^{(\lambda)}_{\ell+1} V_\lambda.$$

For $\lambda = (3, 2, 1^q)$, $n = q + 5$ and $\ell = q + 2$, so $\Omega = R'_{q+3} R'_{q+4} R'_{q+5}$ has three R'-blocks. By the May-13 multi-corner-dimension paper, $\dim B = f^{\lambda^{(\geq 2)}} = f^{(2,1)} = 2$ in the stable regime $q \geq 3$.

Seminormal support. For $W \subseteq V_\lambda$, define

$$\operatorname{supp}(W) \;:=\; \{\, T \in \mathrm{SYT}(\lambda) : \text{some } w \in W \text{ has } [v_T]\, w \neq 0 \,\},$$

i.e., the union of seminormal supports of vectors in W.

Column-1 identity prefix. For an SYT T of any partition μ, the *identity prefix length* is

$$p(T) \;:=\; \max\{\, P \geq 0 : T(i,1) = i \text{ for all } 1 \leq i \leq P \,\}.$$

By construction $T(1,1) = 1$, so $p(T) \geq 1$ for any SYT (when μ has at least one cell).

2 Input from prior papers

2.1 The support rule (May-14 second pass)

Proposition 1 (Support rule)**.** *Let $W \subseteq V_\lambda$ be any subspace. Fix $1 \leq j \leq n-1$. If every $T \in \operatorname{supp}(W)$ has j and $j+1$ in the same column of T, then $W \subseteq E_j^- |_{V_\lambda}$.*

Proof. Let $w \in W$ and write $w = \sum_T c_T v_T$ over $T \in \operatorname{supp}(W)$. By hypothesis, every T in this sum has $j, j+1$ in the same column, so by the Hoefsmit same-column rule $T_j v_T = -v_T$ for every contributing T. Linearity gives $T_j w = -w$. □ □

Remark 2. If every $T \in \operatorname{supp}(W)$ has identity-prefix length $p(T) \geq P$, then j and $j+1$ both sit at column-1 rows $j, j+1$ for every $j \leq P-1$ (so they share column 1). Hence $W \subseteq E_j^- |_{V_\lambda}$ for $1 \leq j \leq P-1$.

2.2 The May-20 three-branch reduction

For $\lambda = (3, 2, 1^{n-5})$ at $n \geq 8$, the May-20 paper `2026-05-20-3-2-1n-family.tex` establishes a structural decomposition of B along corner pairs. Recall λ has three removable corners $c_1 = (1,3)$, $c_2 = (2,2)$, $c_3 = (n-3, 1)$; the three doubly-stripped partitions are

$$\mu_A = \lambda \setminus \{c_1, c_3\} = (2, 2, 1^{n-6}), \qquad \mu_B = \lambda \setminus \{c_2, c_3\} = (3, 1^{n-5}), \qquad \mu_C = \lambda \setminus \{c_1, c_2\} = (2, 1^{n-4}),$$

each of size $n-2$. For each $X \in \{A, B\}$ (the μ_C piece will be irrelevant for us), the May-20 paper constructs an injective $H_q(S_{n-2})$-equivariant linear map $\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$ as follows. Write $\{c_X^-, c_X^+\}$ for the corresponding ordered pair of corners ($c_A^\pm = (c_1, c_3)$, $c_B^\pm = (c_2, c_3)$); for each $T' \in \mathrm{SYT}(\mu_X)$, define two SYTs T_A^X, T_B^X of λ by extending T' with $\{n-1, n\}$ at $\{c_X^-, c_X^+\}$ in the two possible orders, and set

$$\Phi_X(v_{T'}) \;:=\; v_{T_A^X} + \tau_X\, v_{T_B^X},$$

where $\tau_X = \tau(d_X) \in \mathbb{Q}(q)^\times$ is the Hoefsmit T_{n-1}-block coefficient at axial distance $d_X = c(c_X^+) - c(c_X^-)$ ($|d_X| \geq 2$ for $n \geq 8$).

Theorem 3 (Three-branch reduction, May-20)**.** *For $\lambda = (3, 2, 1^q)$ with $q \geq 3$ (i.e., $n \geq 8$),*

$$B \;=\; (T_{n-3}{+}1)(T_{n-2}{+}1)\Big(\Phi_A\big(B_{n-3}^{(\mu_A)} V_{\mu_A}\big) + \Phi_B\big(B_{n-3}^{(\mu_B)} V_{\mu_B}\big)\Big). \tag{1}$$

Proof. This is Theorem 3 (and Lemma 4 on the C-piece vanishing) of the May-20 paper, restricted to its first conclusion (the C-piece $\Phi_C(B_{n-3}^{(\mu_C)} V_{\mu_C})$ is shown there to vanish, so it contributes nothing to B). □ □

The May-20 proof uses the abstract Phase A endpoint $R'_n V_\lambda = E^+_{n-1} V_\lambda$, the E^+-decomposition $E^+_{n-1} V_\lambda = \bigoplus_X \Phi_X(V_{\mu_X})$ along the three corner pairs, and a Φ_X-commutation argument (Steps 2–3 of its Theorem 3) to push the remaining R'-factors through Φ_X.

2.3 Equivariance and E^-_j-preservation properties of Φ_X

Lemma 4 (Φ_X is T_j-equivariant for $j \le n-3$, May-20 Lemma 3.2)**.** *For each $X \in \{A, B, C\}$, every $1 \le i \le n-3$, and every $v \in V_{\mu_X}$, $T_i \Phi_X(v) = \Phi_X(T_i v)$. Consequently, for each $1 \le j \le n-3$,*

$$\Phi_X\Big(E^-_j|_{V_{\mu_X}}\Big) \;\subseteq\; E^-_j|_{V_\lambda} .$$

Proof. The equivariance is May-20 Lemma 3.2 (a verbatim adaptation of the May-15 Φ-equivariance argument: T_i for $i \le n-3$ commutes with T_{n-1}, so the construction of Φ_X as a T_{n-1} $+q$-eigenvector inside a 4-dim subspace decouples). The E^-_j-preservation is immediate: if $T_j v = -v$ then $T_j \Phi_X(v) = \Phi_X(T_j v) = -\Phi_X(v)$. □ □

2.4 The $(2, 2, 1^{q'})$ inner piece

Theorem 5 (Extended sign-kill for $(2, 2, 1^{q'})$, May-14 second pass)**.** *For $\lambda' = (2, 2, 1^{q'}) \vdash n' = q'+4$ with $q' \ge 2$, the spanning vector b' of $B^{(\lambda')}_{q'+3} V_{\lambda'}$ has seminormal support*

$$\operatorname{supp}(b') \;=\; \{\, T'^{(a,c)} : (a, c) \in \mathcal{S}_{n'} \,\}, \qquad \mathcal{S}_{n'} = \{(n'-3, n'-2), (n'-3, n'-1), (n'-2, n'), (n'-1, n')\},$$

where $T'^{(a,c)}$ has $T'(1, 2) = a$, $T'(2, 2) = c$, and column 1 filled with $\{1, \ldots, n'\} \setminus \{a, c\}$ in increasing order. Every $T' \in \operatorname{supp}(b')$ has identity-prefix length $p(T') \ge a - 1 \ge n' - 4$.

Proof. The support description is Theorem 1 of May-14 second pass (`2026-05-14-extended-sign-kill-22.tex`, citing the May-13 non-hook paper); the prefix bound is Lemma 2 of the same paper: $T'^{(a,c)}(i, 1) = i$ iff $i \le a-1$, and $a \ge n'-3$ over the support, so $a - 1 \ge n' - 4$. □ □

For us $\lambda' = \mu_A$ (so $n' = n-2$, $q' = n-6$), giving $p(T') \ge n-6$ for every $T' \in \operatorname{supp}(B^{(\mu_A)}_{n-3} V_{\mu_A})$.

3 The hook support lemma

We now prove the new ingredient: a column-1 identity-prefix bound for the seminormal support of $B^{(\mu)} V_\mu$ at any column-1 hook $\mu = (k, 1^m)$ in its rank-zero regime $m \ge k$. This will give the analogous bound for the Φ_B-piece.

Theorem 6 (Hook support lemma)**.** *Let $\mu = (k, 1^m) \vdash n_0 := k + m$ with $k \ge 2$ and $m \ge k$. Set $\ell_0 := \ell(\mu) = m + 1$ and $j_0 := \ell_0 + 1 = m + 2$. Then every $T \in \operatorname{supp}(B^{(\mu)}_{j_0} V_\mu)$ satisfies*

$$p(T) \;\ge\; m - k + 2.$$

Proof. By strong induction on $k \ge 2$.

Base case $k = 2$. $\mu = (2, 1^m)$ with $|\mu| = n_0 = m+2$, $\ell_0 = m+1$, $j_0 = m+2 = n_0$. Then $B^{(\mu)}_{j_0} V_\mu = R'_{n_0} V_\mu$. By Phase A (May-19 endpoint), $R'_{n_0} V_\mu = E^+_{n_0-1}|_{V_\mu}$.

The SYTs of $\mu = (2, 1^m)$ are parametrized by $a := T(1, 2) \in \{2, 3, \ldots, n_0\}$; column 1 holds $\{1, \ldots, n_0\} \setminus \{a\}$ in increasing order at rows $1, \ldots, n_0 - 1$. For each T_a (the SYT with $T_a(1, 2) = a$), entries $n_0 - 1, n_0$ sit at:

- $a = n_0$: n_0 at $(1,2)$, n_0-1 at $(n_0-1,1)$. Different row, different column. Hoefsmit 2-block.
- $a = n_0-1$: n_0-1 at $(1,2)$, n_0 at $(n_0-1,1)$. Different row, different column. T_{n_0-1}-partner of the above (swap $\{n_0-1,n_0\}$).
- $a \le n_0-2$: n_0-1, n_0 both in column 1 at rows n_0-2, n_0-1 (by counting). Same column. $T_{n_0-1}v_{T_a} = -v_{T_a}$, so $v_{T_a} \in E^-_{n_0-1}$.

Hence $E^+_{n_0-1}|_{V_\mu}$ is 1-dimensional, spanned by the T_{n_0-1} $+q$-eigenvector $v_{T_{n_0-1}} + \tau\, v_{T_{n_0}}$ of the 2-block $\{T_{n_0-1}, T_{n_0}\}$, where $\tau \neq 0$.

The support consists of exactly two SYTs, T_{n_0-1} and T_{n_0}, with a-values n_0-1 and n_0. Both satisfy $p(T_a) = a-1 \ge n_0-2 = m$. Since $m-k+2 = m$ for $k=2$, the bound is exactly m.

Inductive step $k \ge 3$. Assume the theorem for $k-1$ (i.e., for any hook $\mu' = (k-1, 1^{m'})$ with $m' \ge k-1$). Let $\mu = (k, 1^m)$ with $m \ge k$. Set $\mu_1 := (k-1, 1^{m-1})$, of size $n_0 - 2 = k+m-2$. Note μ_1's parameters $(k-1, m-1)$ satisfy $m-1 \ge k-1$ since $m \ge k$, so the inductive hypothesis applies.

By the May-13 shift-lemma paper (`2026-05-13-shift-lemma-all-hooks.tex`, Corollary 5 formula[1]), the same-row S-piece dies and the Φ_*-piece survives unchanged through the outer factors:

$$B^{(\mu)}_{j_0} V_\mu \;=\; (T_{\ell_0}{+}1)(T_{\ell_0+1}{+}1)\cdots(T_{n_0-2}{+}1)\,\Phi_*\big(B^{(\mu_1)}_{j_0(\mu_1)} V_{\mu_1}\big), \tag{2}$$

where $\Phi_* : V_{\mu_1} \hookrightarrow V_\mu$ is the T_{n_0-1} $+q$-eigenvector embedding at the corner pair $\{c_1, c_*\} = \{(1,k), (\ell_0, 1)\}$ (May-13 r1 Definition 2.1).

Step 1: prefix bound on the Φ_*-image. Let $T' \in \operatorname{supp}(B^{(\mu_1)}_{j_0(\mu_1)} V_{\mu_1})$. By the inductive hypothesis (with $k_1 = k-1$, $m_1 = m-1$), $p(T') \ge m_1 - k_1 + 2 = m-k+2$.

Now $\Phi_*(v_{T'})$ is a linear combination of v_{T_A}, v_{T_B} where T_A, T_B are the two SYTs of μ obtained from T' by placing $\{n_0-1, n_0\}$ at $\{(1,k), (m+1,1)\}$ in the two possible orders. In both, the cells of μ_1's column 1 (namely rows $1, 2, \ldots, m$) carry the same entries as in T':

$$T_A(i,1) = T_B(i,1) = T'(i,1) \quad \text{for } 1 \le i \le m.$$

(The added cell $(1,k)$ is in row 1 and the added cell $(m+1,1)$ is at the bottom of column 1, so neither alters rows $1, \ldots, m$ of column 1.)

By IH, $T'(i,1) = i$ for $i \le m-k+2$, hence the same holds for T_A, T_B:

$$p(T_A) \ge m-k+2, \qquad p(T_B) \ge m-k+2.$$

Therefore every SYT in $\operatorname{supp}(\Phi_*(B^{(\mu_1)}_{j_0(\mu_1)} V_{\mu_1}))$ has identity-prefix length $\ge m-k+2$.

Step 2: outer factors preserve the prefix. We use the following elementary lemma (proved in Section 3.1):

Lemma 7 (Outer-factor preservation)**.** *Let $W \subseteq V_\nu$ be a subspace such that every $T \in \operatorname{supp}(W)$ satisfies $p(T) \ge P$. Then for every j with $j \ge P+1$, every $T'' \in \operatorname{supp}((T_j{+}1)W)$ also satisfies $p(T'') \ge P$.*

The factors in (2) are $(T_j{+}1)$ for $j \in \{\ell_0, \ell_0+1, \ldots, n_0-2\} = \{m+1, m+2, \ldots, n_0-2\}$. We have $j \ge m+1$. The required prefix bound is $P = m-k+2$, so $P+1 = m-k+3$. The condition $j \ge P+1$ becomes $j \ge m-k+3$, which is implied by $j \ge m+1$ since $k \ge 2$.

[1] Equivalent to the dim-bound proof: from Phase A plus the same-row-dies argument plus the Φ_*-commutation, one extracts that the entire image equals $(T_{\ell_0}{+}1)\cdots(T_{n_0-2}{+}1)\,\Phi_*\big(B^{(\mu_1)}_{j_0(\mu_1)} V_{\mu_1}\big)$.

Applying Lemma 7 sequentially with each factor, the prefix bound $\geq m-k+2$ is preserved through every outer factor.

Conclusion. By (2), every $T'' \in \operatorname{supp}(B^{(\mu)}_{j_0} V_\mu)$ has $p(T'') \geq m-k+2$, completing the induction. □ □

3.1 Proof of Lemma 7

Proof. Take any $w \in W$, written $w = \sum_T c_T v_T$ over $T \in \operatorname{supp}(W)$. For each T, the Hoefsmit action gives $(T_j+1)v_T \in \operatorname{span}(v_T, v_{s_jT})$, where s_jT denotes the SYT obtained from T by swapping the cells of entries j and $j+1$ (well-defined as an SYT iff $j, j+1$ lie in different rows and columns; otherwise s_jT is undefined and the action gives either qv_T or $-v_T$, with no off-diagonal contribution).

We claim: if $p(T) \geq P$ and $j \geq P+1$, then s_jT (when defined) also satisfies $p(s_jT) \geq P$.

Indeed, $p(T) \geq P$ means $T(i,1) = i$ for $1 \leq i \leq P$; equivalently, the entries $\{1, 2, \ldots, P\}$ occupy precisely the column-1 cells $\{(1,1), (2,1), \ldots, (P,1)\}$. The swap s_j moves only entries j and $j+1$. Since $j \geq P+1 > P$, neither j nor $j+1$ is in $\{1, \ldots, P\}$, so the cells $\{(1,1), \ldots, (P,1)\}$ remain occupied (in order) by entries $\{1, \ldots, P\}$ after the swap. Hence $(s_jT)(i,1) = i$ for $1 \leq i \leq P$, i.e., $p(s_jT) \geq P$.

Since $\operatorname{supp}((T_j+1)v_T) \subseteq \{T, s_jT\}$ (or $\{T\}$ alone when s_jT undefined), and both sides of this set have prefix $\geq P$, we conclude $\operatorname{supp}((T_j+1)w) \subseteq \bigcup_T \operatorname{supp}((T_j+1)v_T)$ has every member with prefix $\geq P$. □ □

Remark 8. Lemma 7 is purely a property of the T_j action and the column-1 structure of an SYT. It does not require any specific shape; it applies to any partition ν. This universality is exactly what lets us combine prefix bounds across the two inner pieces and outer factors of equation (1).

4 Main theorem

Theorem 9 (Main)**.** *For $\lambda = (3, 2, 1^q) \vdash n = q+5$ with $q \geq 3$, the image $B = B^{(\lambda)}_{\ell+1} V_\lambda$ satisfies*

$$B \subseteq \bigcap_{j=1}^{q-2} E_j^- |_{V_\lambda} .$$

Proof. By Remark 2, it suffices to show that every $T \in \operatorname{supp}(B)$ has $p(T) \geq q-1 = n-6$. We use the three-branch reduction (1) together with the prefix bounds on the two inner pieces and Lemma 7 on the outer factors.

Step 1: prefix bound on the Φ_A-image. By Theorem 5 applied to $\mu_A = (2, 2, 1^{n-6})$ (so $n' = n-2$, $q' = n-6 \geq 2$), every $T' \in \operatorname{supp}(B^{(\mu_A)}_{n-3} V_{\mu_A})$ has $p(T') \geq n'-4 = n-6$.

Now $\Phi_A(v_{T'})$ is supported on two SYTs T^A_A, T^A_B obtained from T' by placing $\{n-1, n\}$ at $\{c_1, c_3\} = \{(1,3), (n-3,1)\}$ in the two possible orders. Cells of μ_A's column 1 are at rows $1, 2, \ldots, \ell(\mu_A) = n-4$. The added cell $(1,3)$ is in row 1 but column 3 (not column 1), and the added cell $(n-3, 1)$ is at row $n-3 > n-4$, i.e., *below* μ_A's column 1. Hence rows $1, 2, \ldots, n-4$ of column 1 in T^A_A, T^A_B are unchanged from T':

$$T^A_A(i,1) = T^A_B(i,1) = T'(i,1) \quad \text{for } 1 \leq i \leq n-4.$$

By the prefix bound on T', $T'(i,1) = i$ for $i \leq n-6$, hence $p(T^A_A), p(T^A_B) \geq n-6$.

Step 2: prefix bound on the Φ_B-image. By Theorem 6 applied to $\mu_B = (3, 1^{n-5})$ (so $k = 3$, $m = n - 5 \geq 3$ for $n \geq 8$, i.e., $q \geq 3$), every $T' \in \operatorname{supp}(B_{n-3}^{(\mu_B)} V_{\mu_B})$ has $p(T') \geq m - k + 2 = n - 6$.

Now $\Phi_B(v_{T'})$ is supported on two SYTs T_A^B, T_B^B obtained from T' by placing $\{n-1, n\}$ at $\{c_2, c_3\} = \{(2,2), (n-3,1)\}$ in the two possible orders. Cells of μ_B's column 1 are at rows $1, 2, \ldots, \ell(\mu_B) = n - 4$. The added cell $(2,2)$ is in column 2 (not column 1), and $(n-3, 1)$ is below μ_B's column 1. As in Step 1, rows $1, \ldots, n-4$ of column 1 in T_A^B, T_B^B agree with T', so $p(T_A^B), p(T_B^B) \geq n - 6$.

Step 3: outer factors preserve the prefix. The outer factors in (1) are $(T_{n-3} + 1)$ and $(T_{n-2} + 1)$. Both indices $j \in \{n-3, n-2\}$ satisfy $j \geq n - 3 = (n-6) + 3 \geq (n-6) + 1$, so the hypothesis $j \geq P + 1$ of Lemma 7 (with $P = n - 6$) is met. Applying the lemma twice, the prefix bound $p \geq n - 6$ is preserved.

Conclusion. By (1),

$$B \;=\; (T_{n-3} + 1)(T_{n-2} + 1)\left(\Phi_A(\cdots) + \Phi_B(\cdots)\right),$$

and we have just shown that the two inner sums have supports with prefix $\geq n - 6$, and the outer factors preserve this. Therefore every $T \in \operatorname{supp}(B)$ has $p(T) \geq n - 6$, and by Remark 2, $B \subseteq E_j^-$ for $1 \leq j \leq n - 7 = q - 2$. $\square$ $\square$

5 Sharpness

We separate sharpness into a structural reduction (Lemma 10, fully rigorous) and a residual computational input (Theorem 11, verified for $q \in \{3, 4, 5, 6\}$).

Lemma 10 (Sharpness reduction)**.** *Suppose $B \subseteq E_{q-1}^- |_{V_\lambda}$. Then every $T \in \operatorname{supp}(B)$ satisfies $p(T) \geq n - 5$.*

Proof. Set $j_0 := q - 1 = n - 6$. Suppose for contradiction $T \in \operatorname{supp}(B)$ has $p(T) = n - 6$ exactly. Since $T \in \operatorname{supp}(B)$ there is $w \in B$ with $C_T := [v_T]\, w \neq 0$, and by hypothesis $T_{j_0}\, w = -w$.

Locating entry $n-5$ in T. By $p(T) \geq n-6$, the cells $\{(1,1), (2,1), \ldots, (n-6, 1)\}$ are filled with entries $\{1, 2, \ldots, n-6\}$, and the remaining six cells $\{(1,2), (1,3), (2,2), (n-5,1), (n-4,1), (n-3,1)\}$ hold entries $\{n-5, n-4, n-3, n-2, n-1, n\}$. Since $p(T) = n - 6$ exactly, $T(n-5, 1) \neq n - 5$. We claim entry $n - 5$ is forced to sit at $(1,2)$:

- Not at $(n-4, 1)$ or $(n-3, 1)$: column-increase forces $T(n-5, 1) < n-5$, but no eligible entry remains.
- Not at $(1,3)$: row-increase forces $T(1,2) < n - 5$, but $T(1,2) \in \{n-4, \ldots, n\}$ (all entries $\geq n - 4 > n - 5$ fail).
- Not at $(2,2)$: column-2 increase forces $T(1,2) < T(2,2) = n - 5$, but $T(1,2)$ would have to equal one of $\{n-4, n-3, n-2, n-1, n\}$, all $> n - 5$.

Hence $T(1,2) = n - 5$.

Off-diagonal swap. Cells of $j_0 = n-6$ at $(n-6, 1)$ and $j_0 + 1 = n-5$ at $(1,2)$ are in different rows ($n - 6 \neq 1$ for $n \geq 8$) and different columns; axial distance $|d| = |c(1,2) - c(n-6,1)| = |1 - (7-n)| = n - 6 \geq 2$ for $q \geq 3$. Hoefsmit 2-block: $T_{j_0}\, v_T = \alpha\, v_T + \beta\, v_{T^*}$ with $\beta \neq 0$, where $T^* := s_{j_0} T$ swaps cells of $n-6$ and $n-5$ (verified to be a valid SYT: column 1 at row $n-6$ now holds $n-5 < T(n-5,1)$, and column 2 at row 1 holds $n-6 < T(2,2)$ since $T(2,2) \in \{n-4, \ldots, n\}$).

Prefix drop forces $T^* \notin \mathrm{supp}(B)$. $T^*(n-6,1) = n-5 \neq n-6$, so $p(T^*) \leq n-7 < n-6$. By Theorem 9, $T^* \notin \mathrm{supp}(B) \supseteq \mathrm{supp}(w)$.

Coefficient contradiction. Computing $[v_{T^*}]\,(T_{j_0}\,w) = \sum_{U\in\mathrm{supp}(w)} C_U \cdot [v_{T^*}]\,(T_{j_0}\,v_U)$. The diagonal contribution $[v_{T^*}]\,(T_{j_0}\,v_{T^*})$ requires $T^* \in \mathrm{supp}(w)$ (no). The off-diagonal contribution $[v_{T^*}]\,(T_{j_0}\,v_U)$ for $U \neq T^*$ requires $s_{j_0}U = T^*$, equivalently $U = s_{j_0}T^* = T$. Hence

$$[v_{T^*}]\,(T_{j_0}\,w) \;=\; C_T \cdot \beta \;\neq\; 0, \quad \text{while} \quad [v_{T^*}]\,(-w) \;=\; 0.$$

This contradicts $T_{j_0}\,w = -w$. We conclude every $T \in \mathrm{supp}(B)$ satisfies $p(T) \geq n-5$. □ □

Theorem 11 (Sharpness, computationally verified)**.** *For $\lambda = (3,2,1^q)$ at every $q \in \{3,4,5,6\}$, $\mathrm{supp}(B)$ contains an SYT T with $p(T) = n-6$ exactly; consequently $B \not\subseteq E^-_{q-1}|_{V_\lambda}$.*

Proof. Existence of $T \in \mathrm{supp}(B)$ with $p(T) = n-6$ is verified computationally for $q \in \{3,4,5,6\}$ (Theorem 15); explicit witness at $q = 3$ ($n = 8$): $T \in \mathrm{SYT}(\lambda)$ with $T(1,1) = 1$, $T(1,2) = 3$, $T(1,3) = 5$, $T(2,1) = 2$, $T(2,2) = 7$, $T(3,1) = 4$, $T(4,1) = 6$, $T(5,1) = 8$ (column-1 entries $(1,2,4,6,8)$, prefix $= 2 = n-6$, with $\{n-1,n\} = \{7,8\}$ at the corner pair $\{c_2,c_3\} = \{(2,2),(5,1)\}$ — i.e., T lies in Φ_B's image extension). Combining with Lemma 10, the assumption $B \subseteq E^-_{q-1}$ forces $\mathrm{supp}(B)$ to omit such T, contradicting existence. Hence $B \not\subseteq E^-_{q-1}$. □ □

Remark 12 (Toward fully structural sharpness)**.** A fully structural sharpness proof would require showing *structurally* that $\mathrm{supp}(B)$ achieves $p(T) = n-6$ at every $q \geq 3$. The witness above (for $q = 3$) is the T^B_A-extension under Φ_B of an inner support SYT of $B^{(\mu_B)}$ with column-1 entries $(1,2,m,m+2)$ ($m = n-5$). The structural obstacle is two-fold: (i) the May-12 "hook Catalan-support" conjecture (all support coefficients of b'_B nonzero) is presently empirical, and (ii) outer-factor cancellations $(T_{n-3}{+}1)(T_{n-2}{+}1)$ may in principle annihilate specific Φ_X-image SYTs (and computationally do reduce $|\mathrm{supp}(\Phi_A(b'_A){+}\Phi_B(b'_B))|$ from at most $|\mathcal{S}_{\mu_A}|{\cdot}2{+}|\mathrm{supp}(B^{(\mu_B)})|{\cdot}2 = 18$ to $|\mathrm{supp}(B)| = 26$ — no, the converse: outer factors can both kill and create supports; empirically the support *grows* from the inner support). A direct dimension argument would also work: $\dim B = 2$ and the inclusion $B \subseteq E^+_\ell \cap \bigcap_{j=1}^{q-1} E^-_j$ (under the contrary-to-sharpness assumption) would give B inside a specific ≤ 2-dimensional subspace which can be computed. We leave this for a future cycle.

Remark 13 (Stronger empirical statement)**.** Computational verification at $q \in \{3,4,5,6\}$ (mod $P = 100\,003$, $Q = 11$) shows that $S_{j_0}|_B$ has full rank $2 = \dim B$ at $j_0 = q-1$, i.e., $B \cap E^-_{j_0} = 0$ (not merely $B \not\subseteq E^-_{j_0}$). Theorem 11 proves the weaker non-containment.

6 Eigenspace summary

Combined with the universal $B \subseteq E^+_\ell$ (May-13 evening Theorem A), Theorem 9 and Theorem 11 give the following complete picture for the T_j-eigenspace containments of B in the family $\lambda = (3,2,1^q)$, $q \geq 3$:

Theorem 14 (Eigenspace summary)**.** *For $\lambda = (3,2,1^q)$ with $q \geq 3$,*

$$B \;\subseteq\; E^+_\ell \;\cap\; \bigcap_{j=1}^{q-2} E^-_j\,|_{V_\lambda}, \qquad B \not\subseteq E^-_{q-1}|_{V_\lambda}\;.$$

The E^+_ℓ containment is sharp in the sense that $T_\ell|_B = q\cdot\mathrm{Id}_B$ (and so $B \cap E^-_\ell = 0$).

Proof. The E_j^- containment for $j \leq q-2$ is Theorem 9; the non-containment at $j = q-1$ is Theorem 11; the E_ℓ^+ containment with the scalar action is May-13 evening Theorem A. $\square$ $\square$

The eigenspace constraints alone do *not* pin down B. Computationally $\dim\big(E_\ell^+ \cap \bigcap_{j=1}^{q-2} E_j^-|_{V_\lambda}\big) \gg \dim B = 2$ at every q tested. The dimension formula $\dim B = f^{\lambda^{(\geq 2)}} = f^{(2,1)} = 2$ (May-13 multi-corner-dimension) is a sharper fact.

7 Computational verification

Theorem 15 (Verification)**.** *Theorem 14 and Theorem 6 were verified computationally at $q \in \{3,4,5,6\}$ (i.e., $n \in \{8,9,10,11\}$) and at hook parameters $(k,m) \in \{(2,2),\ldots,(2,6),(3,3),\ldots,(3,7),(4,4),\ldots,(4,8$ mod $P = 100\,003$ at $Q = 11$. In every case:*

- *$\dim B = f^{\lambda^{(\geq 2)}}$, matching the May-13 multi-corner formula.*
- *$|\mathrm{supp}(B)| = 26$ for $\lambda = (3,2,1^q)$, independent of q.*
- *Every $T \in \mathrm{supp}(B)$ has $p(T) \geq q-1 = n-6$, with the bound achieved (i.e., the minimum is exactly $n-6$).*
- *$T_j b = -b$ for every $b \in B$ and $1 \leq j \leq q-2$.*
- *$S_{q-1}|_B$ has rank $\dim B = 2$ (sharper than Theorem 11; cf. Remark 13).*
- *For hook $\mu = (k,1^m)$ with $m \geq k$: $|\mathrm{supp}(B^{(\mu)}_{j_0(\mu)} V_\mu)| = C_{k-1}$ (the $(k-1)$-th Catalan number, $1,2,5,14,42,\ldots$), and the minimum identity-prefix length is exactly $m-k+2$.*

Proof. Scripts in `~/projects/scratch/2026-05-14-extended-sign-kill-32/` and `~/projects/scratch/2026-(` The scripts construct H_q-Hoefsmit matrices, build Ω and the inner $\Omega^{(\mu_X)}$ explicitly, compute supports by row-echelon reduction of the image-basis matrix, and compute S_j-rank by direct linear algebra. See in particular `probe_hook_general.py` for the hook lemma data and `explore_32_family.py` for the $(3,2,1^q)$ data. $\square$ $\square$

8 Discussion

8.1 What is new

The extended sign-kill conjecture (May-14 evening) asserts, for $\lambda = (\widehat{\lambda}, 1^q)$ with $\ell(\widehat{\lambda}) \geq 2$,

$$B \subseteq E_j^-|_{V_\lambda} \iff j \leq \tau(\lambda), \qquad \tau(\lambda) = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2.$$

Prior to this paper:

- May-13 `2026-05-13-non-hook-22-1n.tex`: $j = 1$ for $\widehat{\lambda} = (2,2)$.
- May-13 `2026-05-20-3-2-1n-family.tex`: $j = 1$ for $\widehat{\lambda} = (3,2)$.
- May-14 second pass `2026-05-14-extended-sign-kill-22.tex`: full chain $j \leq q-1$ for $\widehat{\lambda} = (2,2)$ (containment + sharpness).

This paper closes the $\widehat{\lambda} = (3,2)$ row in the containment direction at the predicted threshold $\tau = q-2$, and proves sharpness at the boundary $j = q-1$. The remaining $j \in \{q, q+1, \ldots, n-1\} \setminus \{\ell\}$ non-containments are not addressed; computational data agrees with non-containment throughout this range.

8.2 The hook support lemma as a stand-alone result

Theorem 6 is, to my knowledge, the first clean support description for hook j-formulas at general arm length k. The May-12 hook paper conjectured a Catalan-many support of size C_{k-1}; the present paper does not characterize the support set, but the identity-prefix bound $p(T) \geq m-k+2$ that it does prove is exactly what the May-13 shift-lemma induction makes available. Empirically the support *equals* the set of SYTs with $p \geq m-k+2$ *and* $\{n_0-1, n_0\}$ at certain positions; characterizing this exactly would close the May-12 Catalan-support conjecture and is a natural follow-up.

8.3 Why the top-two-rows independence holds (partial answer)

The formula $\tau(\lambda) = q+3-\widehat{\lambda}_1-\widehat{\lambda}_2$ depends only on the first two rows of $\widehat{\lambda}$. For $\widehat{\lambda} = (2,2)$ the proof reduces directly to the May-13 four-SYT support; for $\widehat{\lambda} = (3,2)$ the proof reduces via three-branch to a $\mu_A = (2,2,1^{n-6})$ piece *and* a $\mu_B = (3,1^{n-5})$ piece. Both inner pieces have the *same* prefix bound $n-6$:

- For $\mu_A = (2,2,1^{n-6})$, the bound $a-1 \geq n'-4 = n-6$ comes from the explicit support description.
- For $\mu_B = (3,1^{n-5})$ (a hook), the bound $m-k+2 = n-6$ comes from the shift-lemma induction, with the inductive step carrying the bound through the Φ_*-extension and outer factors unchanged.

Both bounds equal $n-6$ *because* of the coincidence $\widehat{\lambda}_1+\widehat{\lambda}_2-2 = 3$ in this family. This suggests the top-two-rows independence is a consequence of: (i) every doubly-stripped μ_X for an $r=2$ multi-corner shape inherits the same prefix bound, and (ii) outer factors and Φ_X-extensions preserve the bound. A precise general statement is left for future work.

8.4 Empirical extension to general $\widehat{\lambda}$

Computational verification (mod $P = 100\,003$ at $Q = 11$) at $\widehat{\lambda} \in \{(4,2),(5,2),(3,3),(4,3)\}$ across 20+ test shapes (May-14 evening session, also confirmed in May-14 second-pass paper Section 6.3) shows the same prefix bound $p(T) \geq n-\widehat{\lambda}_1-\widehat{\lambda}_2$ in every tested case. The natural conjecture:

Conjecture 16 (Support prefix bound, general $\widehat{\lambda}$)**.** *For $\lambda = (\widehat{\lambda}, 1^q)$ with $\ell(\widehat{\lambda}) \geq 2$ and q in the stable regime, every $T \in \operatorname{supp}(B^{(\lambda)}_{\ell+1}V_\lambda)$ satisfies $p(T) \geq q+3-\widehat{\lambda}_1-\widehat{\lambda}_2$.*

By Remark 2, Conjecture 16 would imply the "$\subseteq$" direction of the full extended sign-kill conjecture for general $\widehat{\lambda}$.

8.5 Containment versus identity

Theorem 14 gives $B \subseteq E^+_\ell \cap \bigcap_{j=1}^{q-2} E^-_j$. The intersection on the right is computationally much larger than $\dim B = 2$ in every tested case; the eigenspace constraints alone do not pin down B. The dim formula $f^{\lambda(\geq 2)} = 2$ (May-13 multi-corner) is a sharper fact, capturing structure not detected by T_j-eigenspaces.

8.6 Status of the broader iso programme

This result closes the $\widehat{\lambda} = (3,2)$ row of the May-14 evening conjecture in the containment direction but does not affect the broader iso programme (the May-13–May-14 attempts to identify B as a known representation-theoretic object, all closed off by the May-14 morning, afternoon, and evening refutations). The present contribution is, in effect, a continued sharpening of the empirical extended sign-kill into a structurally established theorem at the next non-trivial $\widehat{\lambda}$.

9 Status

Established.

- Theorem 6: hook support prefix bound (new).
- Theorem 9: $B \subseteq E_j^-$ for $j \leq q-2$ in the $\widehat{\lambda} = (3,2)$ family, $q \geq 3$.
- Theorem 11: sharpness at $j = q-1$.
- Theorem 14: complete eigenspace summary.

Open.

- Sharpness for $j \in \{q, q+1, \dots, n-1\} \setminus \{\ell\}$ (consistent computationally; structural argument analogous to May-14 second-pass paper Proposition 10 should adapt).
- Conjecture 16 for general $\widehat{\lambda}$ (containment direction of the full extended sign-kill conjecture).
- Closed-form characterization of $\operatorname{supp}(B^{(\mu)})$ for hooks (May-12 Catalan conjecture).

Push status. TBD — depends on PAT refresh. Prior unpushed-commit count on `clio-vega/proofs`: 65 (May-14 fourth-pass session).

Scripts.

- `~/projects/scratch/2026-05-14-extended-sign-kill-proof/explore_32_family.py` — support and E_j^- ranks for $(3,2,1^q)$.
- `~/projects/scratch/2026-05-14-extended-sign-kill-32/probe_hook_general.py` — hook prefix-lemma verification across (k,m) table.